# L2.2 The Mean Value Theorem

Work in your group. Check your answers against the Quarto tonight.

## Rolle's Theorem

**1.** Verify that the function satisfies the three hypotheses of Rolle's Theorem on the given interval. Then find all numbers $c$ that satisfy the conclusion.

$$f(x) = \sin\left(\tfrac{x}{2}\right) \qquad \text{on } \left[\tfrac{\pi}{2}, \tfrac{3\pi}{2}\right]$$

> **Stop.** Check the hypotheses one at a time, and only then use the conclusion.

**2.** Rolle's Theorem does not apply to three of these. For each one, say whether it applies, and if it does not, name **which hypothesis fails**.

(a) $f(x) = x^2 - 6x + 8$ on $[1, 5]$

(b) $f(x) = |x - 2|$ on $[0, 4]$

(c) $f(x) = \dfrac{1}{x - 1}$ on $[0, 2]$

(d) $f(x) = x^2$ on $[0, 3]$

**3.** Let $f(x) = \tan x$. Then $f(0) = f(\pi)$, but there is no number $c$ in $(0, \pi)$ with $f'(c) = 0$. **Why does this not contradict Rolle's Theorem?**

## The Mean Value Theorem

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**4.** Verify that the function satisfies the hypotheses of the Mean Value Theorem on the given interval. Then find all numbers $c$ that satisfy the conclusion.

$$f(x) = 2x^2 - 3x + 1 \qquad \text{on } [0, 2]$$

**5.** Verify that the function satisfies the hypotheses of the Mean Value Theorem on the given interval. Then find all numbers $c$ that satisfy the conclusion.

$$f(x) = \frac{1}{x} \qquad \text{on } [1, 3]$$

**6.** $f(x) = x^3 - 3x^2$ on $[0, 3]$. Find every $c$ that satisfies the conclusion of the Mean Value Theorem.

**7.** Let $f(x) = 2 - |2x - 1|$. Show that there is no value of $c$ with $f(3) - f(0) = f'(c)(3 - 0)$. **Why does this not contradict the Mean Value Theorem?**

## Exactly one real root

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**8.** Show that the equation $2x - 1 - \sin x = 0$ has **exactly one** real root.

**9.** Suppose $f$ is differentiable everywhere and has two roots. Show that $f'$ has at least one root.

## Lawyer Math

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**10.** BOARDS **Group notes.** For each theorem below: write the statement **exactly**, list its hypotheses one per line, and then do the problem beside it.

⑤ **Rolle's Theorem.** Verify that $f(x) = x^3 - 2x^2 - 4x + 2$ satisfies the three hypotheses on $[-2, 2]$, then find all numbers $c$ that satisfy the conclusion.

⑥ **Mean Value Theorem.** Verify that $f(x) = x^3 - 3x + 2$ satisfies the hypotheses on $[-2, 2]$, then find all numbers $c$ that satisfy the conclusion.

With these two the set of six is complete. **On your board, put all six in order** and mark each one as *existence*, *location*, or *procedure*.

① IVT ② EVT ③ Fermat ④ Closed Interval Method ⑤ Rolle ⑥ MVT